

DP Manager User Guide: Frequently Asked Questions (3rd Edition) Nov. 30, 2020

For questions or comments, contact the support team for the product under which this component is invoked.

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1. Preface

About This Document

The DP Manager User Guide: Frequently Asked Questions provides a brief overview of CDPL and a detailed description of how to use its “DP Manager” user interface. It shows you how to use the DP Manager to monitor, start and control a distributed process run.

Scope

The document aims to answer frequently asked questions about using the DP Manager GUI. It is however not an authority on CDPL or distributed programming as such. Refer to related publications for information about CDPL.

ACRONYM	DESCRIPTION
CDPL	Common Distributed Processing Library
DP	Distributed Processing
GUI	Graphical User Interface
PID	Process Identifier
SGE	Oracle Grid Engine (formerly Sun Grid Engine)
RSH	Remote Shell
LSF	Load Sharing Facility
RTDA	Runtime Design Automation
PBS	Portable Batch System
SLURM	Simple Linux Utility for Resource Management

Related Publications

For additional information about CDPL, see

- DP Manager User Guide
- CDPL User’s Manual¹

¹ Currently only available to Synopsys R&D and field application engineers.

2. What Is CDPL?

The Common Distributed Processing Library (CDPL) is a common framework used by Synopsys applications to incorporate parallel and distributed computing.

In parallel and distributed computing, a single job is divided into smaller tasks which are then executed concurrently. It can be used effectively to meet the increased demands of certain EDA applications for computation, performance, memory, and data.

Through Synopsys CDPL, users can:

- Launch processes across a distributed set of nodes.
- Schedule the distributed processing work through RSH, SSH, SGE, LSF, PBS, RTDA, NetBatch, SLURM and Shell.
- Track the status of the work and progress of a DP run.
- Change the DP runtime environment with a configuration file.
- Collect run time statistics and system performance data.
- Benefit from automatic job control, detection of errors and recovery from failed tasks

3. What Are the Main Components Of CDPL?

In the CDPL framework, the smaller partitions (i.e., tasks) of an application (e.g., EDA) process are collectively referred to as a **JOB**. The application program that creates, submits, and starts the JOB is called the **MASTER**. The application program that accepts and processes the smaller tasks is called the **WORKER**. The MASTER creates the job and tasks and dispatches the tasks to appropriate WORKERS. It handles all required communication between itself and the WORKERS. As the MASTER and WORKERS execute their functions, CDPL has a mechanism to query the state of the DP entities (MASTER, WORKER, JOB, TASK, etc.) and the state of the processing nodes (host status information). This is made possible through an entity called a **MONITOR**. The MONITOR is the interface between the DP run and any reporting front ends that need to be built on top of CDPL. By default, CDPL ships with the **DP MANAGER** application that uses the monitor.

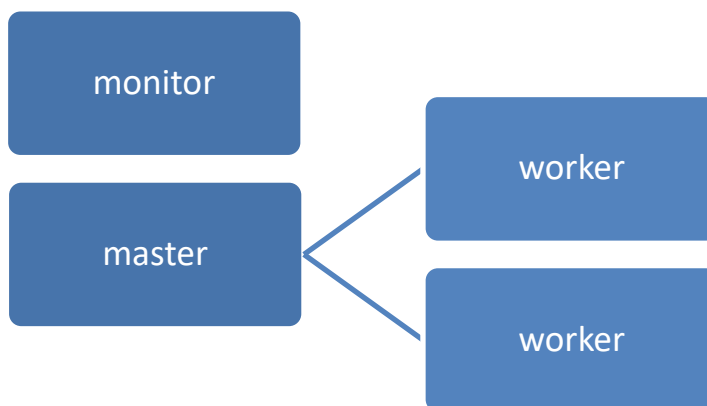


Figure 1

4. What Would I Use the DP Manager For?

The DP MANAGER can monitor all activities performed by the CDPL and provides limited interactive operations to control a few other events:

USE CASE	DP MANAGER COMPONENT
JOB/TASK MONITORING	DP MONITOR
DEBUGGING	DP LOG BROWSER
DP JOB CONFIGURATION	CONFIGURATION EDITOR
DP EFFICIENCY ANALYSIS	DP ANALYZER
START DP JOB	START DP RUN
HOST MONITORING	HOST STATUS WINDOW
JOB CONTROL	DP MONITOR CONTEXT MENU

5. How Do I Set the Environment for DP Manager?

1. **CDPL_HOME** (required): Set it to the root of the CDPL GUI SDK. The DP Manager uses **\$CDPL_HOME/bin** to locate auxiliary applications such as the “dpanalyzer” and “host_info” scripts.
2. **CDPL_BCASTDIR** (optional): The CDPL Master writes the broadcasts file required by DP Manager to **\$HOME/.synopsys/cdpl** by default. However, you may use **CDPL_BCASTDIR** environment variable to instruct the CDPL Master to write the broadcast files to a different location.

6. How Does the DP Manager Connect to A DP Run?

1. It looks for a broadcast file in the default location **\$HOME/.synopsys/cdpl**. This file contains the coordinates to the CDPL Master. One may use **CDPL_BCASTDIR** environment variable to alter this behavior (as described above).

7. Can the DP Manager Connect to Multiple DP Runs?

1. Yes. On start up the DP Manager scans the default search location for broadcast files. It then connects to the CDPL Master specified in each broadcast file. On successful connection, the DP Manager will show the Job status of each CDPL Run.

8. How Do I Start A DP Run Through the DP Manager?

1. One may start a DP Run using the DP Manager in the following two ways:
 - a. Use the `--run` command line option to pass the executable and its arguments to the DP Manager binary:
Example: `dpmanager --run countcomments SAMPLE_INPUT_1`
One can also pass a startup script to the DP Manager instead:
Example: `dpmanager --run START_DEMO.sh`
 - b. Go to Tools->Start DP Run to select a startup script for execution. One can also use the Start DP run tool button to do the same as shown below:

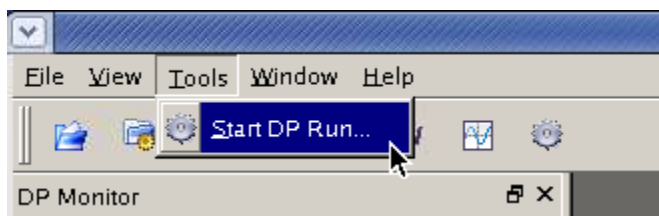


Figure 2

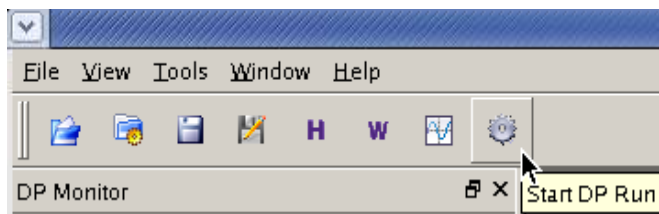


Figure 3

Clicking on either the “Start DP Run” menu item or the tool button, will display a file chooser. Select the startup script from the file chooser and press Open to start your run.

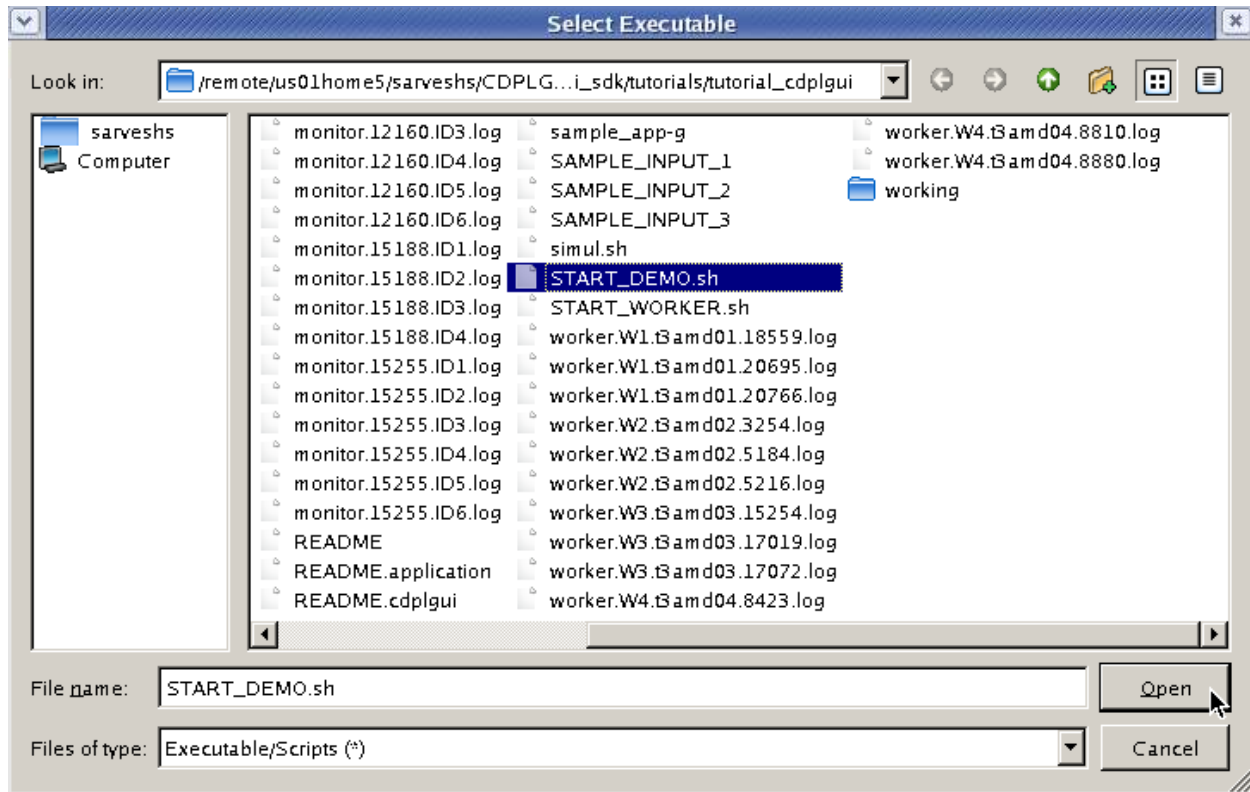


Figure 4

9. Where Can I See the Standard Output/Error Messages Of My DP Run?

1. Standard output and standard error messages can only be seen if you start a DP Run using the DP Manager (see description above).
2. Go to View->DP Process and select the name of your startup script from the sub menu as shown below:

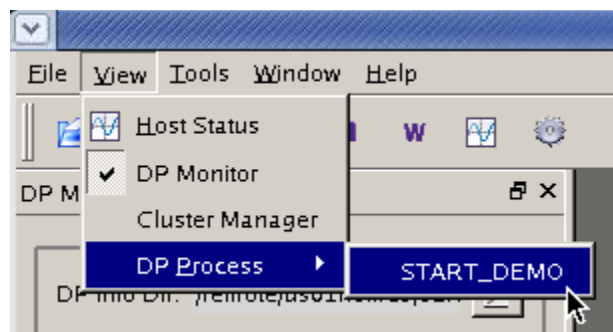


Figure 5

Clicking on the startup script's name (START_DEMO in this example) will display its output window (shown below). The Output window displays the name of the executable, its PID, the full path to the executable and its standard output and error messages.

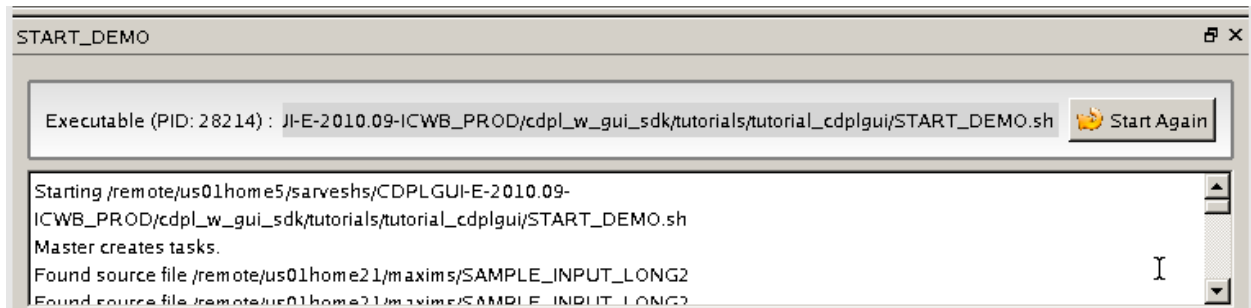


Figure 6

10. What Is The “Start Again” BUTTON in The DP Process Window?

You may use this button to start your DP run again without having to search and select your startup script again. It is activated at the end of the run.

11. What Kind Of Monitoring Information Does The DP Manager Display?

1. The DP Manager provides read access to all activities performed by CDPL. All this information is encapsulated in the “DP Monitor” window.
 - a. Status of running Jobs/Tasks.
 - b. Details about the CDPL Worker (its ID, PID and host).
 - c. Job Summary
 - d. An Interface to the CDPL Master and Worker logs.
 - e. CPU and Memory status of the host machines.
 - f. Information about the DP Info Directory. The DP Info directory is the location of the CDPL broadcast file (usually \$HOME/.synopsys/cdpl).

12. Why Don’t I See Any Monitoring Information in The DP Monitor Window?

1. Check the value of the DP Info Directory displayed in the DP Monitor window. Ensure that it is pointing to the right location. In the unlikely case that it is not pointing to the right location, you can change the DP Info Directory by using the button provided:

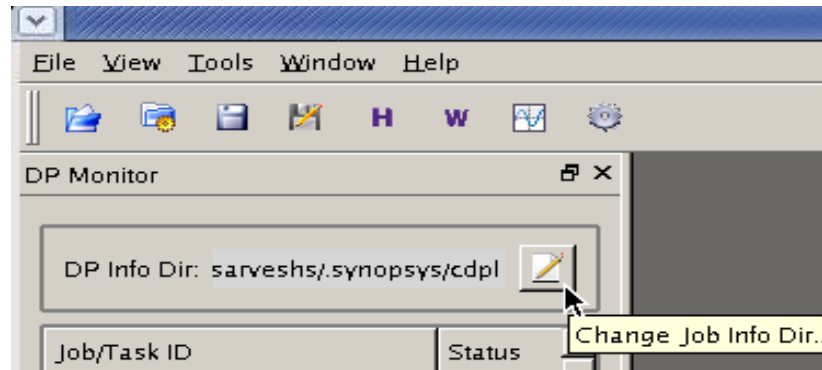


Figure 7

13. How Do Select A Hidden Directory as My DP Info Directory?

1. Click on the “Change Job Info Dir..” button shown above. A “Select Directory” dialog should pop up.
2. Right click on the dialog and choose the “Show hidden files” option from the context menu (shown below).
3. Double click on the required hidden directory and click on choose.

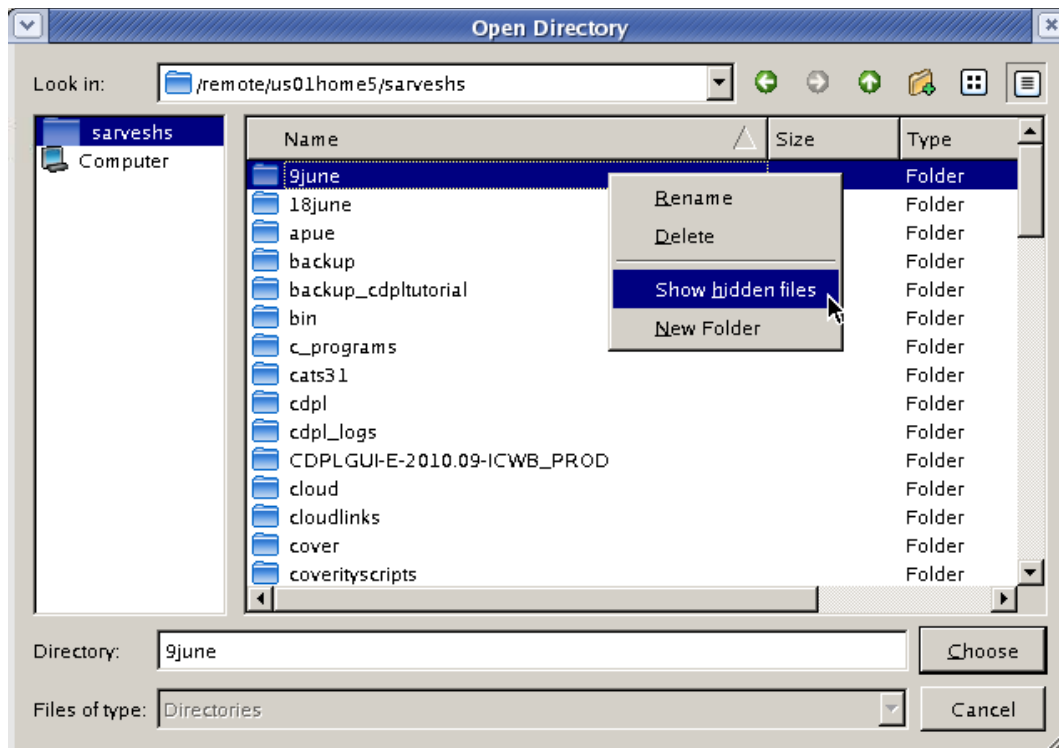


Figure 8

14. How Do I Check the Status of DP Jobs and Tasks?

1. The DP Manager monitors a DP Run for Job-Task status information keeping the user updated about the state of the DP Job, the state of its tasks and other useful information. This information is shown in the DP Monitor Window shown below.

The major states of a job or task can be:

- a. STOPPED (can be restarted)
- b. KILLED (cannot be restarted)
- c. INTERRUPTED (Run was interrupted externally, e.g. Ctrl-C)
- d. RUNNING
- e. COMPLETED
- f. STARTED
- g. SCHEDULED
- h. INPROGRESS
- i. NEW

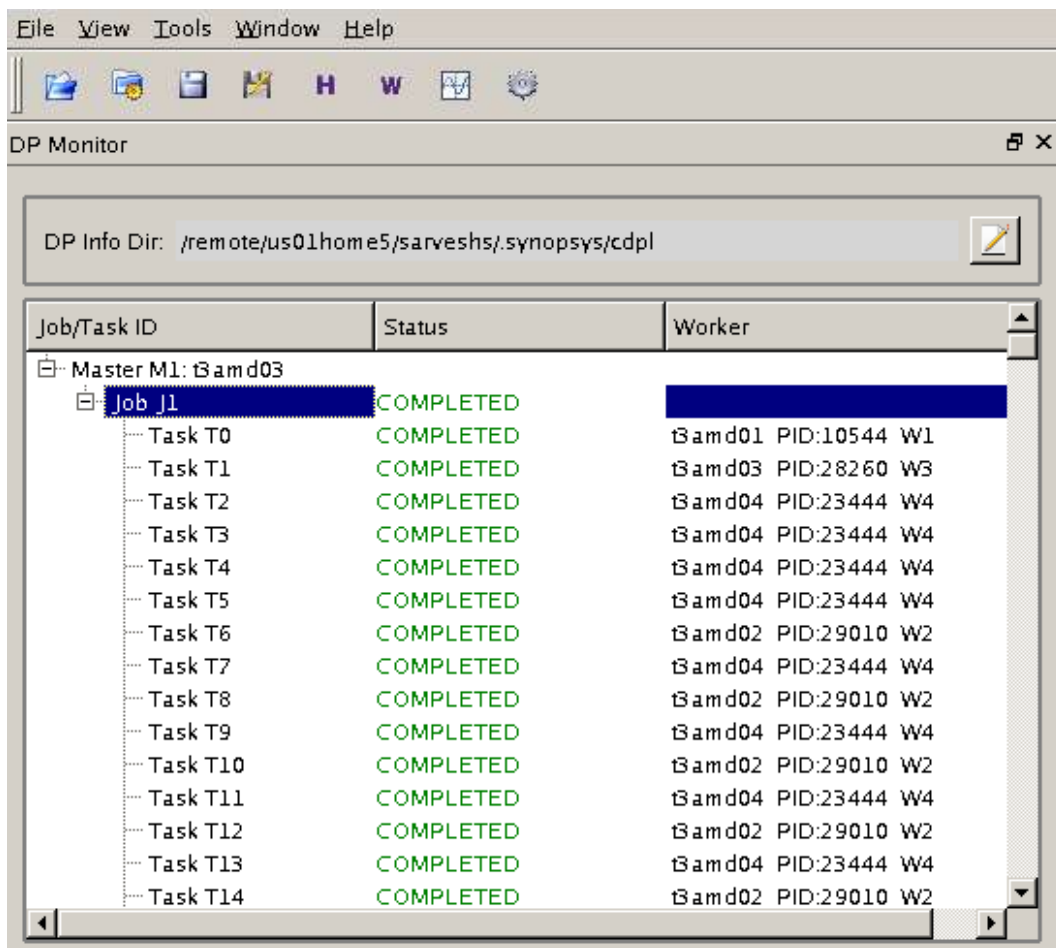


Figure 9

2. The DP Monitor also displays details about the individual Job in the Job Summary section shown below. The Job summary displays the name of the job currently summarized and visualizes its overall progress with the aid of a progress bar.

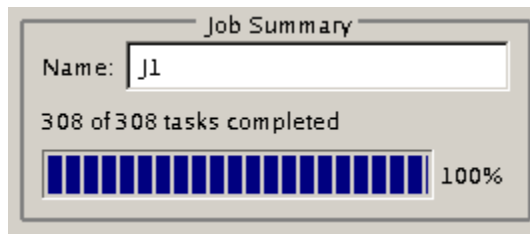


Figure 10

15. How Do I View the Summary of Multiple Jobs in The Above Window?

1. The DP Manager could be connected to several DP Runs or a single DP Run may contain multiple jobs. To view the summary of a Job simply click on its name in the tree below:

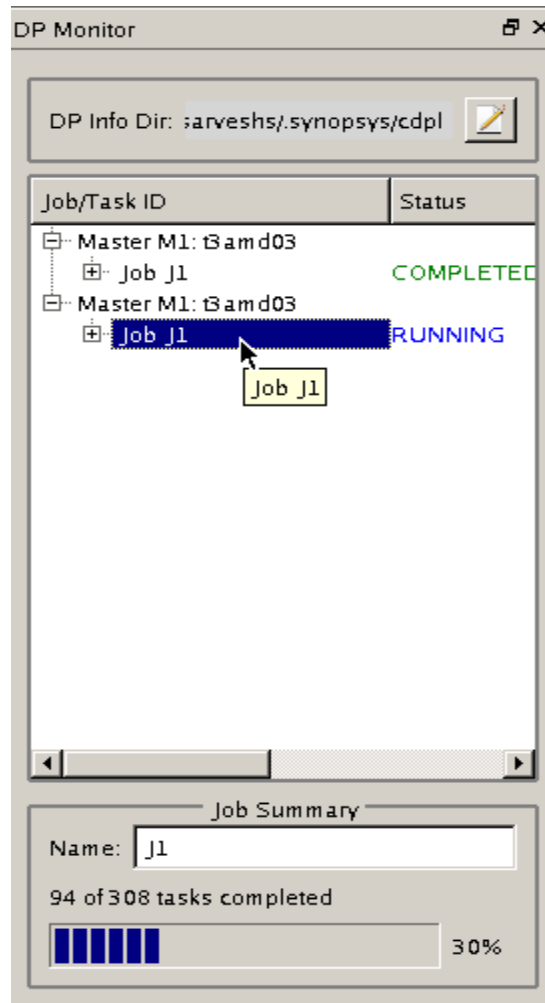


Figure 11

In the above figure, the DP Manager is connected to two DP Runs. Each CDPL Master is added as a top-level item in the job tree. By default, the Job Summary window shows the status of the most current Job. Click on the name of the job from the drop-down tree and view its summary.

16. What Kind of Write Access Does the DP Manager Provide to A Job?

One can use the context menu provided in DP Monitor window (shown above) to:

1. Stop a running Job
2. Kill a running job
3. Start a stopped job

To access the context menu, right click on the name of the Job in the Job tree displayed in DP Monitor window.

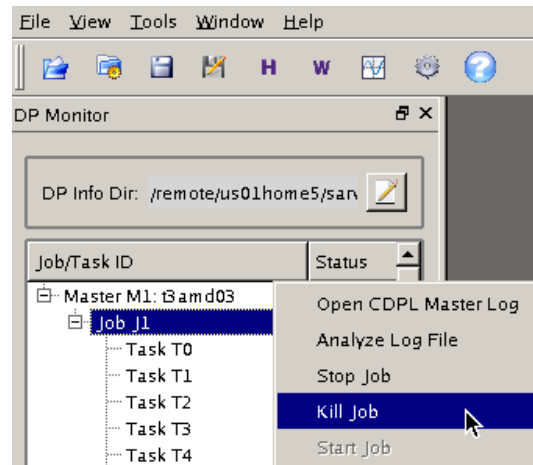


Figure 12

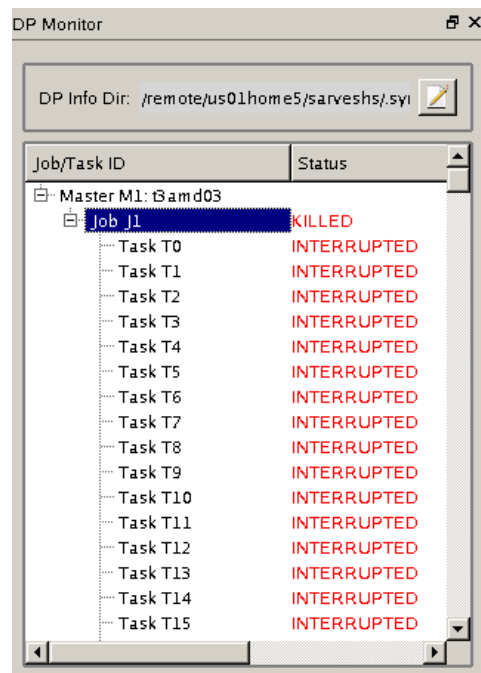


Figure 13

17. Why Does the Stop Job Option Shown Above Not Work for Me?

The application needs to explicitly handle the stop event initiated by DP Monitor. Check with your application support team whether this feature is supported.

18. Why Would I Want to Open DP Log Files Through the DP Manager?

1. Two reasons:
 - a. You need not search for CDPL log files. The DP Manager retains the location of the CDPL Master and Worker log files across runs started in the same session.
 - b. The CDPL log files and particularly the Master log file contains a lot of information as it records all CDPL activities. Isolating important events in the log files can become cumbersome.
To make debugging easier, the DP Manager filters CDPL log files and highlights key events.
2. To view a CDPL Master log file, right click on the name of the Master or on any of its jobs as shown below:

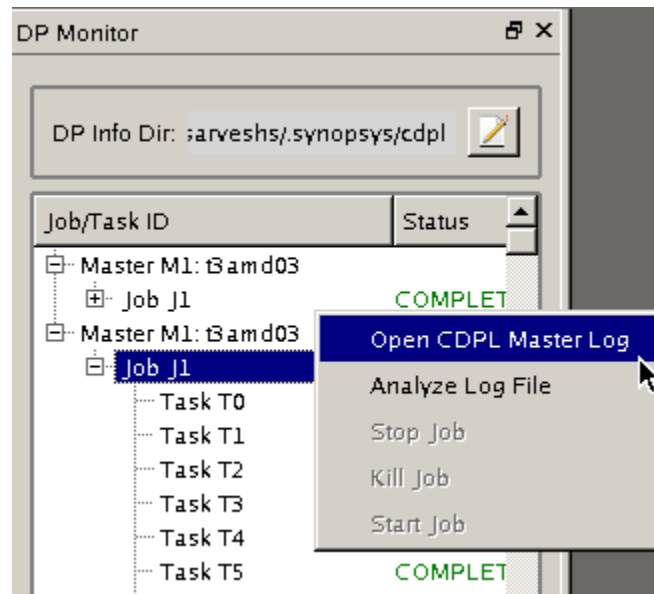


Figure 14

The log file will then be displayed in a new "DP Log Browser" window to the right. One can use the search box provided to isolate keywords.

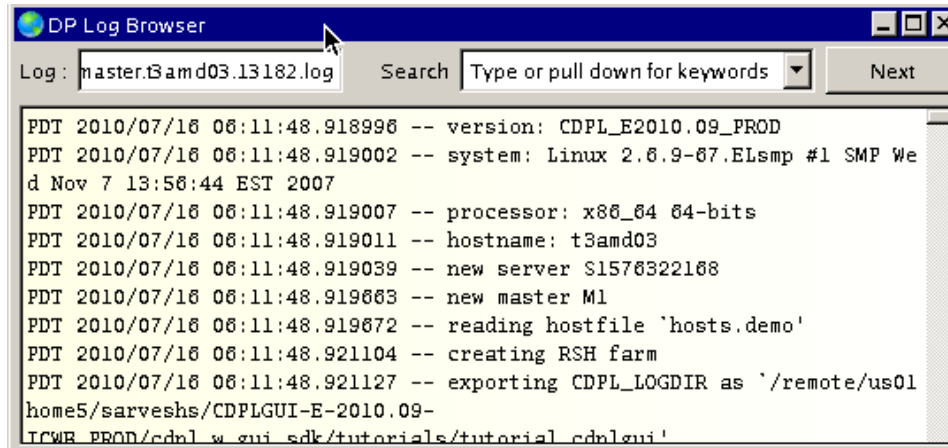


Figure 15

19. How Do I View A DP Worker Log File?

1. Right click on the name of the worker in the Worker column to open the worker log in a new DP Log Browser window as show below:

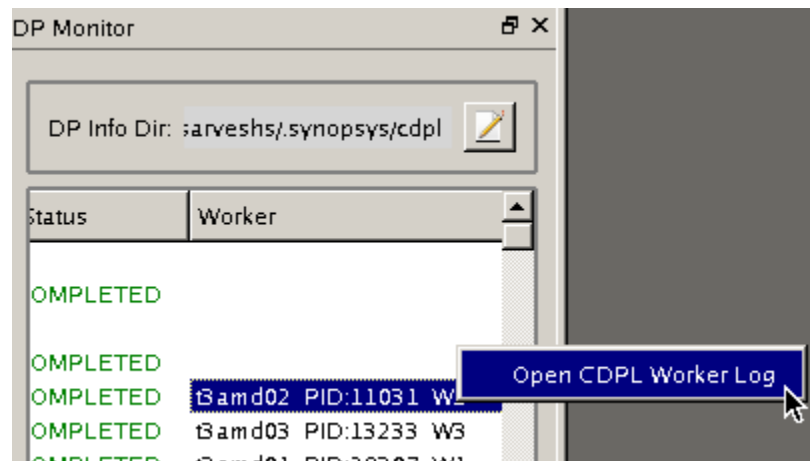


Figure 16

Note: For very short DP Runs (says 5 seconds) the DP Manager may not be able to retrieve all the information required to display the worker log in time. In such cases, only the host name is displayed in the worker column as shown in the figure below. The DP Manager did not get enough time to retrieve the worker ID and its PID. Hence in such cases, you may not be able to right click on the worker name to display its log.

Note: The DP Manager cannot display worker information for the tasks that have already completed when the monitor connects to the job master. The information is not available. The worker column will show "Not available" for these tasks.

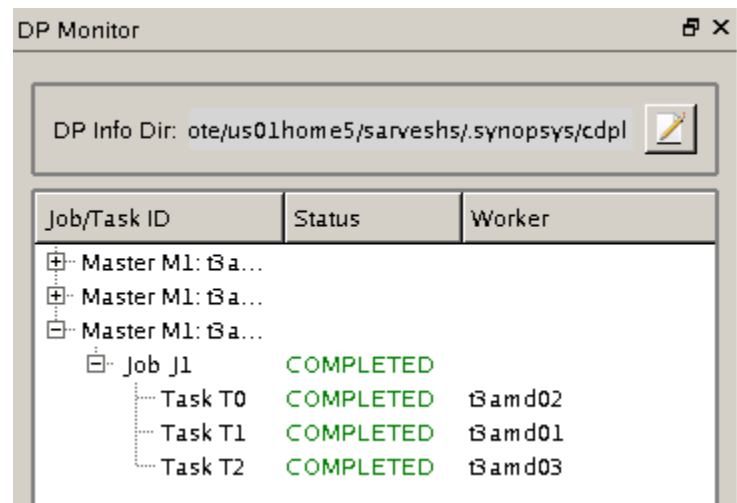


Figure 17

20. Where Is the Status of The Hosts Used in The DP Run Displayed?

1. The DP Manager displays relevant information about the hosts used for the DP Run. Information such as the CPU/Memory utilization, number of workers, current load, etc. are encapsulated in the "Host Status" Window.
2. To View the Host Status window, go to View->Host Status or click on the Host Status icon in the tool bar:

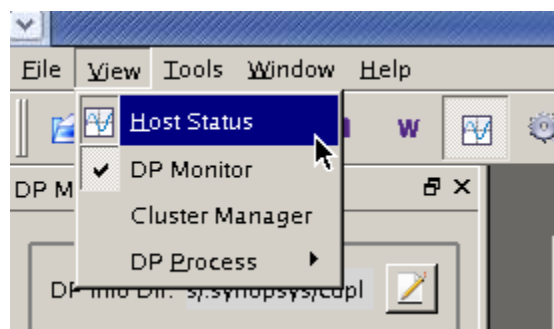


Figure 18

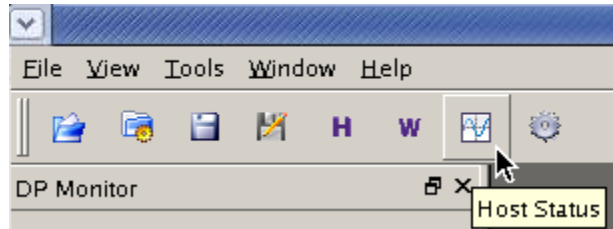


Figure 19

3. The Host Status window displays the CPU, Memory Utilization, Network Status, Disk Status and Load Average in separate tabs. It also shows miscellaneous information about the host such as its current load, current available memory, number of Workers in the host, etc. as a tool tip.

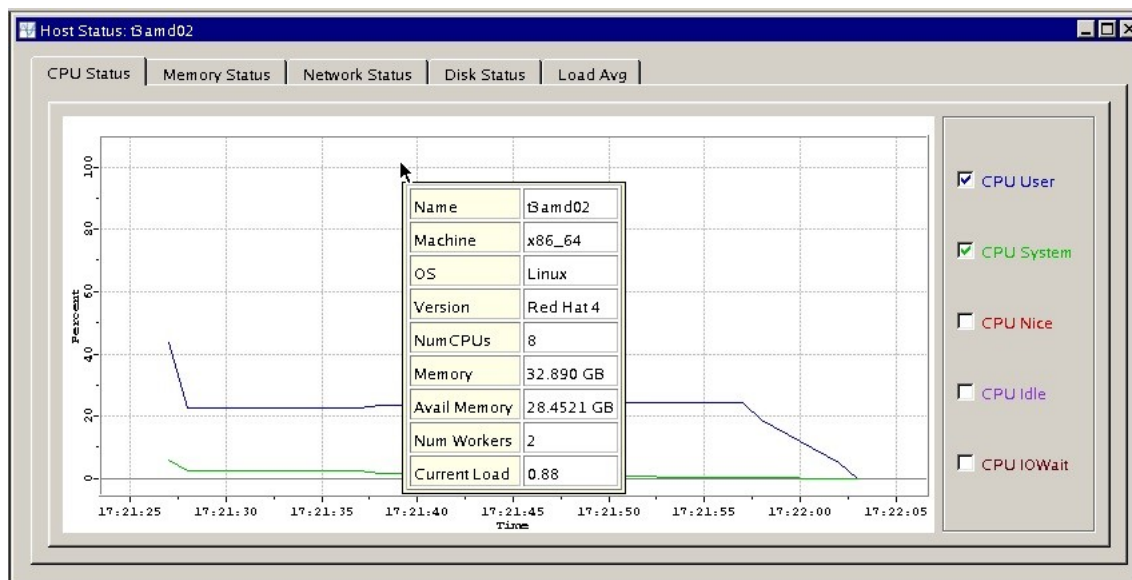


Figure 20

21. How Do I View the Status of Each Host Through the DP Monitor?

1. Click on the name on any row containing the name of a task to see the status of its host. You may also click on the name of a Job to view the aggregate status of all the hosts:

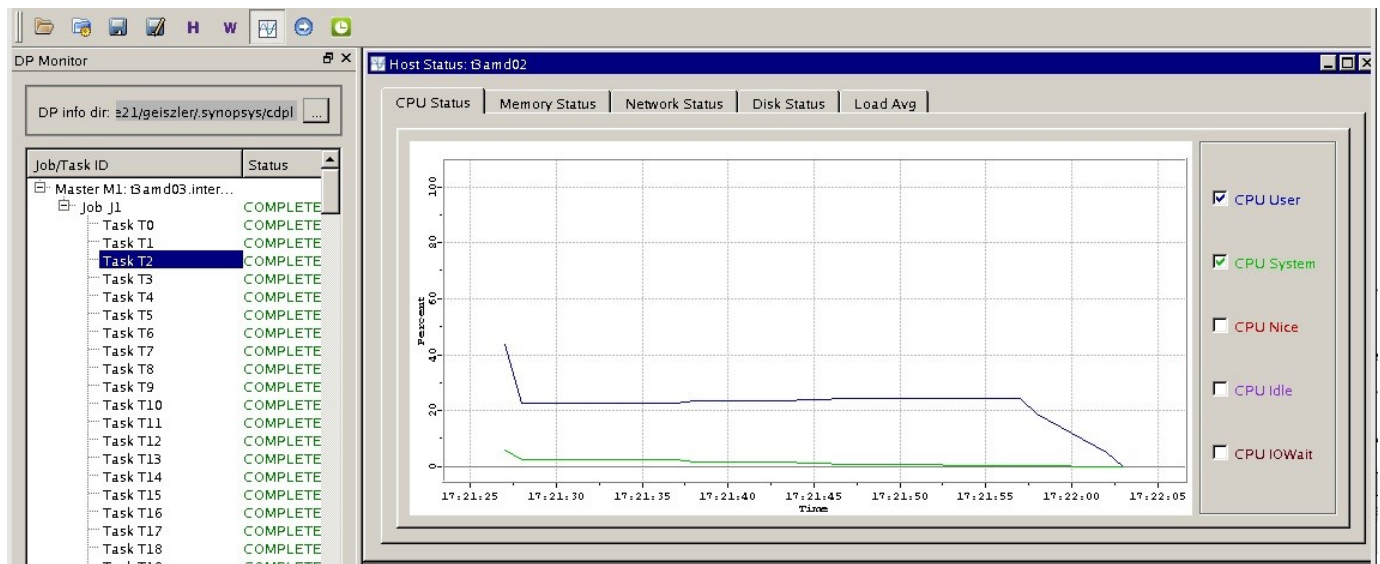


Figure 21

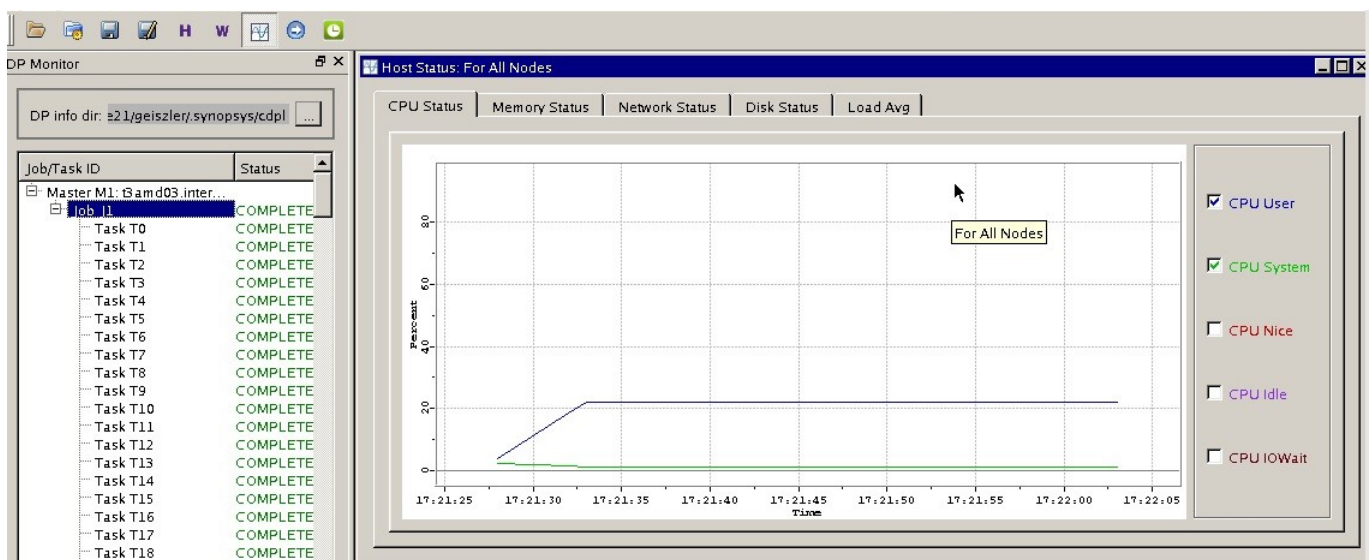


Figure 22

22. How Do I Supply Host and Worker Information to The DP Master?

A distributed run requires interfacing to multiple hosts/nodes to launch WORKERS from a single MASTER application. The DP Manager provides a simple, clean, and uniform interface to these elements.

CDPL supports 7 types of transports that the MASTER can use to dispatch workers on multiple remote nodes:

RSH (remote shell), SSH (secure shell), SH (shell), Platform-LSF (Platform Load Sharing Facility), SGE (Oracle Grid Engine), PBS (Portable Batch System) and RTDA (Runtime Design Automation). Note that Oracle Grid Engine was formerly known as Sun Grid Engine.

Depending on the transport chosen by the end-user, CDPL expects different types of system-specific information. The Hosts information file is a consistent way to communicate information to the DP system for any of the above job dispatch mechanisms.

CDPL can handle only one job-dispatch mechanism per session. For e.g. if “RSH” is the preferred mechanism, the Hosts information file should contain a list of valid hosts. Alternatively, if SGE is the preferred mechanism, the Hosts information file contains the SGE command format and the number of slots (processors) required for the parallel job to execute.

```
# HOSTS INFORMATION FILE (RSH)
# Each valid host definition occupies an entire line.
# Line with '#' as the first character is a comment line.
# Format of each line:
#
#enabled| hostname| slots| tmpDir| mode| command
1| teralab-x | 4 | /remote/users/tmp| RSH | rsh
1| teralab-y | 8 | /remote/users/tmp| RSH | rsh
```

A detailed explanation of the hosts file follows. **This file is typically created and provided by the end user of the application.**

FIELD	VALUE	PURPOSE
1 ('1')	Integer – can be '0' or '1'	A '0' signifies that the host cannot be used, a '1' denotes a usable WORKER machine
2 ('teralab-x')	String, valid hostname	This is the hostname of a valid WORKER machine for RSH and SSH. For SH, SGE, LSF, PBS and RTDA, this field is empty
3 ('4')	Integer – any valid integer	This number is used to denote the number of worker slots on this host (RSH, SSH and SH) or farm (LSF, SGE, PBS and RTDA). -1 indicates unlimited.
4 ('/remote/users/tmp')	String – a valid, writable directory	This directory is used for local data storage for an application, if needed
5 ('RSH')	String – command mode	The string signifies the command-mode used to connect to the WORKER machine. The valid entries RSH, SSH, SH, LSF, SGE, PBS and RTDA.
6 ('rsh')	String – the connection command	The given string denotes the remote command used to connect to the WORKER machine.

An important aspect of the MASTER configuration process is supplying a Worker Information file to the MASTER. The Worker Information file provides the following important information to the MASTER:

- The type of WORKER required by the MASTER; this is an integer
- The WORKER executable name and location

- The environment and configuration to be used by the WORKER

A worker information file may not be required if your application provides the data programmatically within the application. A sample Worker Information file is shown below:

```
#Sample WORKER Information File
#worker_type; max_workers; min_workers; parameterized worker_executable invocation;

100; 2; 1; worker_exe {other parameters}
```

FIELD	DESCRIPTION
100	The type of WORKER for executing the MASTER tasks
2	The maximum number of WORKERS – CDPL attempts to start as many workers it can, up to the maximum specified in this file
1	The minimum number of WORKERS needed for successful operation – it should be at least '1'
worker_exe	The worker executable. The worker executable can be prepended with a wrapper script that can be very useful in providing visibility into worker operation. A sample wrapper script (START_WORKER) is provided in the “tutorials” subdirectory of the CDPL SDK.
Other Parameters	Application specific parameters the worker will need

23. How Do I Create A New Hosts File?

1. To create a new CDPL host file, click on New->Host File or click on the Host File icon in the tool bar:

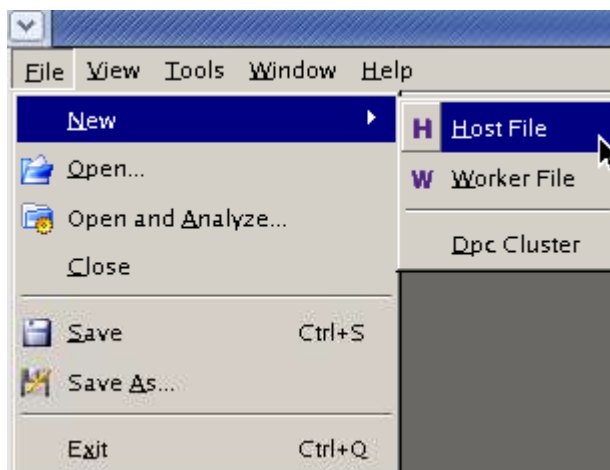


Figure 23



Figure 24

- Specify the command mode (mode used to launch CDPL workers across nodes), temporary directory (optional) and enter the names of the hosts in each row provided by the Configuration Editor Window:

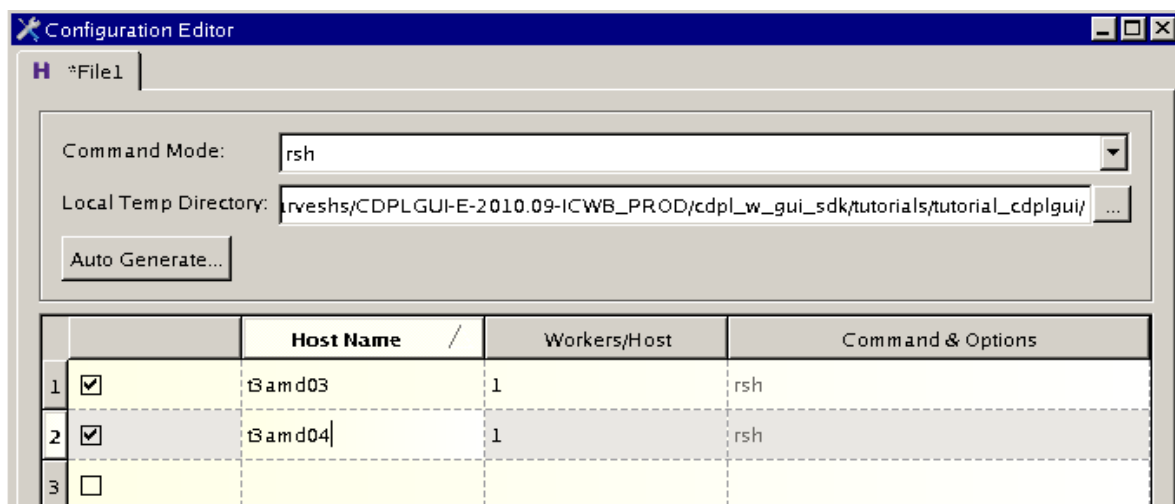


Figure 25

- Save the file to an appropriate directory when you are done.

24. How Do I Create A New Workers File?

- To create a new CDPL workers file, click on New->Worker File or click on the Worker File icon in the tool bar.
- Enter the Worker Type, minimum number of workers, maximum number of workers, select the Worker Executable and specify its input arguments (if any) in the last column:

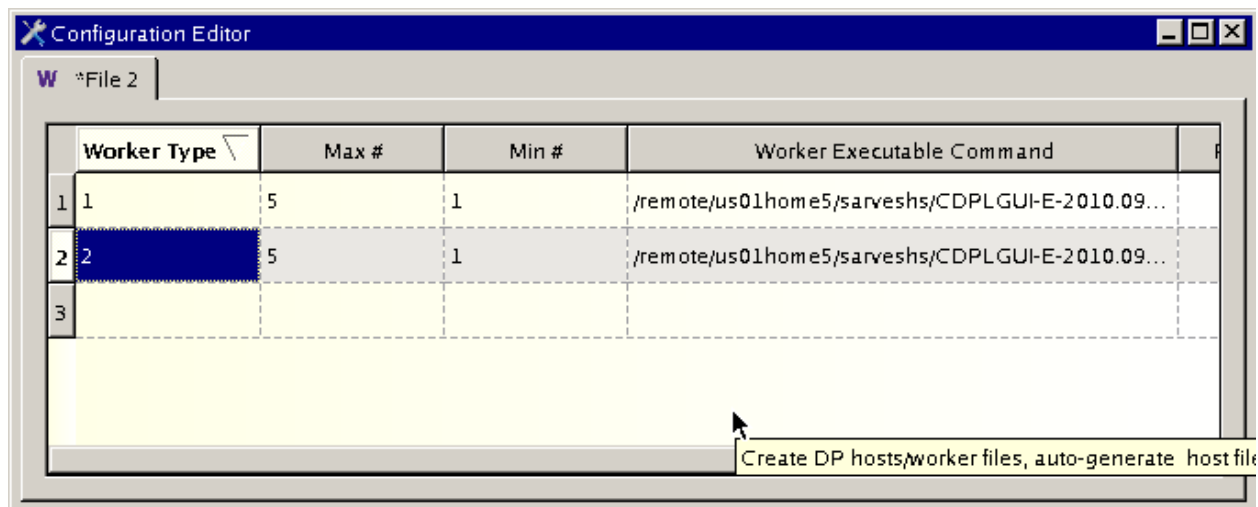


Figure 26

3. Save the File

25. Is There Another Way of Specifying the Hosts and Workers for A DP Run?

1. You may not need DP Manager to specify a list of hosts or to configure the workers for a DP run. Some Synopsys applications may implement their own methods of supplying this data. Check your application documentation for details.

26. What Is the Cluster Manager?

The Cluster Manager provides a user interface that allows users to manage sets of hosts to be used for distributed processing. Each set of hosts is called a cluster. Each cluster has a corresponding CDPL hosts file. The user interface is divided into two views – the Cluster Tree view and Host Status view.

27. How Do I Create A Cluster?

Go to View->Cluster Manager to bring up the Cluster Manager window. Select “DP Clusters” and right-click “New”. A new cluster with an arbitrary name will appear. You can select and edit the cluster name and description to enter your own information about the cluster.

28. How Do I Add Hosts to A Cluster?

Select the cluster in the Cluster Tree View under “DP Clusters”. Right-click and select “Setup”. A host configuration editor window comes up. Enter the host information for your cluster and save the hosts file.

29. Where Is the Cluster Data Stored?

By default, the cluster data is stored in the file `dp_cluster.xml` in the directory `$HOME/.synopsys/cdpl`. `dpcluster.xml` contains the full path name the host file associated with each cluster. The hosts files may be stored anywhere.

30. How Do I View the Status of The Hosts in A Cluster?

Select the cluster in the tree view under “DP Clusters”. The first time you select the cluster; the Cluster Manager will retrieve the status information of each host in the cluster and display it in the Host Status pane. This information will be cached, so that on later selections of the same cluster, the same information will be displayed. You can retrieve and display the status of the hosts in the cluster by selecting the cluster and right-clicking “Refresh”.

31. What Kind of Post-Run Analysis Does the DP Manager Provide?

The primary motivation behind using a DP Scheme is to increase the efficiency of EDA applications. CDPL strives to meet this objective by:

1. Efficient management of computer resources such as CPU's, Task (process) locations
2. Ensuring Optimum Task to CPU mapping
3. Allowing end users to dynamically change worker configuration
4. Managing multiple jobs and job stages with the same resource environment.
5. Minimizing data interchange congestion.
6. Error detection and recovery.

In order to understand the effectiveness of the DP scheme, users would often like to compute this efficiency measure for CDPL.

Efficiency is used to describe properties of an entity relating to how much of various types of resources it consumes. The two most frequently encountered are:

1. Speed or running time
2. Space - the memory or 'non-volatile storage' used

DP Analyzer is the CDPL module that outputs a measure of **total worker occupancy** as efficiency metric.

For example, if the master component was running for 100 seconds, and 10 workers ran an average of 95 seconds executing tasks, the overall efficiency will be 95%. This can be used to roughly estimate the amount of total time that workers were running, but not working on a task. The DP Manager uses DP Analyzer to display this information

You can analyze a log file not visible to the current session of DP Manager (a log file from a past run perhaps) by using the file menu `File->Open and Analyze` or by using the Open and Analyze icon in the tool bar as shown below.

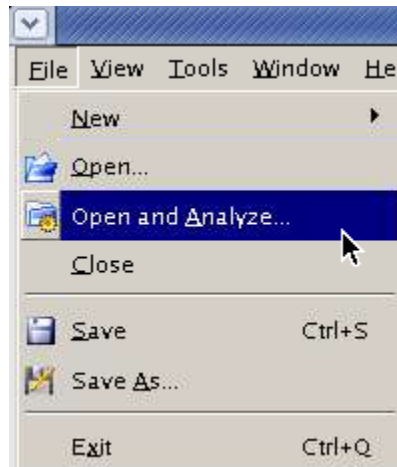


Figure 27

Efficiency = $100 - [(overseer \text{ running time} - average \text{ worker time} / overseer \text{ running time}) * 100]$

Or

Efficiency = $average \text{ worker time} / overseer \text{ running time} * 100$

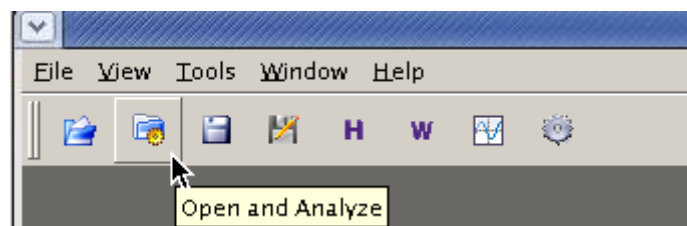


Figure 28

The preferred way of analyzing the Master log relevant to the current session of the DP Manager is to right click on the name of the job and select the “Analyze Log File” option from the context menu.

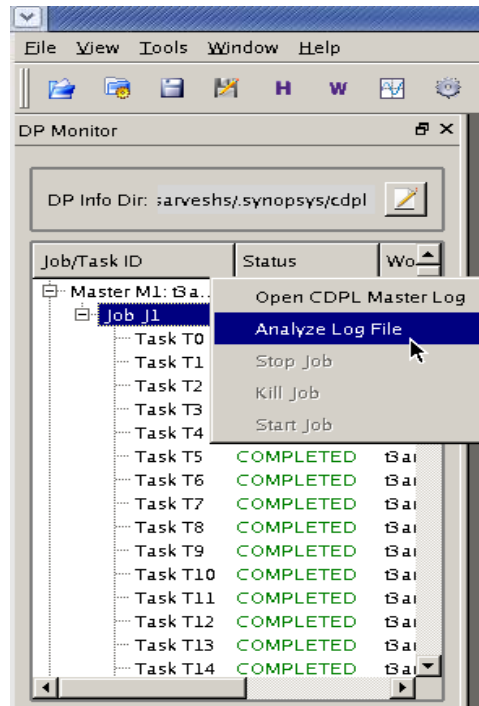


Figure 29

Although primarily meant for post-run analysis, the DP Analyzer may be invoked while the Job is running. If the DP Manager can extract enough useful information from the partially written log file, you may be able to see the DP Analyzer output.

The DP Analyzer shows you a summary of the DP Run in three views: Gantt chart, Tree map and Summary:

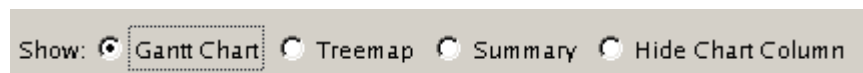


Figure 30

The different views are shown below.

1. Gantt Chart:

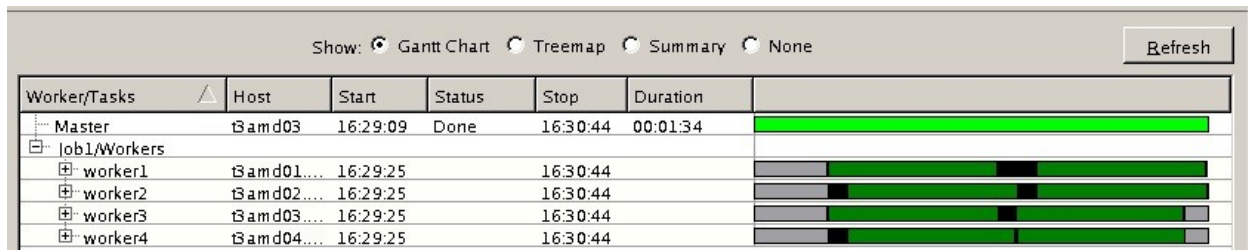


Figure 31

2. Tree Map:

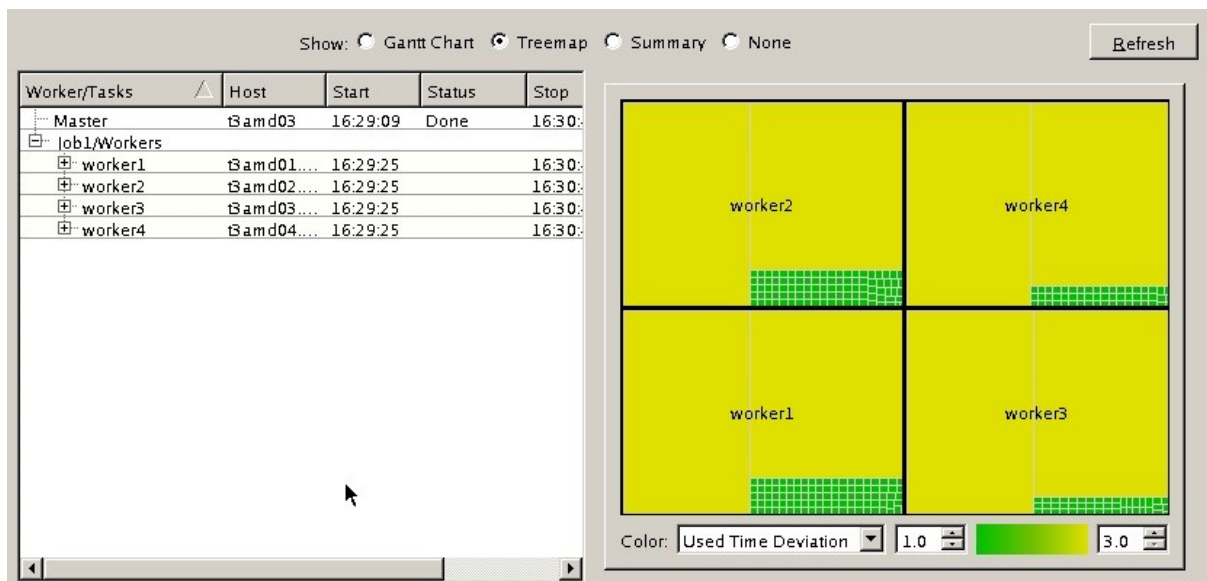


Figure 32

3. Summary highlighting the DP Efficiency (81%):

Job started at: 04:57:59 07/16/2010 PDT
Job stopped at: 04:59:33 07/16/2010 PDT
Total DP job elapsed time: 94 seconds
Workers elapsed time: 94 seconds

Worker	Host	Worked (Sec.)	Num of Tasks	Avg Time (Sec.)
1	t3amd01	78.5	113	0.7
2	t3amd02	78.1	93	0.8
3	t3amd03	73.8	47	1.6
4	t3amd04	73.2	55	1.3

Counting all tasks (no sticher mode)
Total tasks launched: 308
Total tasks completed: 308
Job ran to completion (server shutdown occurred at 94 seconds)
Good= 75.92, Confworkers= 75.92
NFRAC, GOOD = 4,4
All 4 configured workers were useful (completed at least one task)
4 useful workers worked on average 76 of 94 seconds
DP TASK DISTRIB: 81% EFFICIENT
ADDITIONAL POST-WORKER TIME: 0 seconds (unknown cause)
Task times: min/avg/max: 0.04 / 0.99 / 35.96 seconds

Figure 33

32. How Do I See This Manual in DP Manager?

1. Go to Help->DP Manager FAQ



Figure 34

2. This will invoke "acoread" and display this manual in a new window.
3. You will need acoread version 5.0.5 or greater to be installed in your host machine and be present in your path variable.
4. You can display the complete DP Manager User Guide by selecting Help->DP Manager User Guide.